

Range Reversals

Topic.

Square-root decomposition, dynamic arrays, range reversal queries, range sum queries.

Statement.

A physical education teacher has already learned how to compute the total height of all students standing in positions from l to r in a row.

However, the students sometimes play a trick on him. At some moment, the students in positions from l to r swap places. The teacher noticed that the students always reverse this segment: l swaps with r , $l + 1$ swaps with $r - 1$, and so on.

The teacher still wants to compute the total height on all planned segments.

The first line contains two integers n and m :

$$1 \leq n, m \leq 200000,$$

where n is the number of students in the row and m is the number of events.

The second line contains n natural numbers: the heights of the students in their initial order. Each height does not exceed

$$2 \cdot 10^5.$$

Each of the next m lines contains three integers

$$q, l, r,$$

where

$$0 \leq q \leq 1, \quad 1 \leq l \leq r \leq n.$$

If $q = 0$, output the sum of heights on the segment $[l, r]$.

If $q = 1$, reverse the segment $[l, r]$.

For every query of type 0, output one integer: the answer to this query.

Implementation.

The implementation is provided separately in the file

`solution.cpp`.